



Tutorial - Geospatial Foundational Models & Embeddings

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for Earth Observation

Meet the Tutorial Team



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Find any of us if you need any help during the tutorial session!

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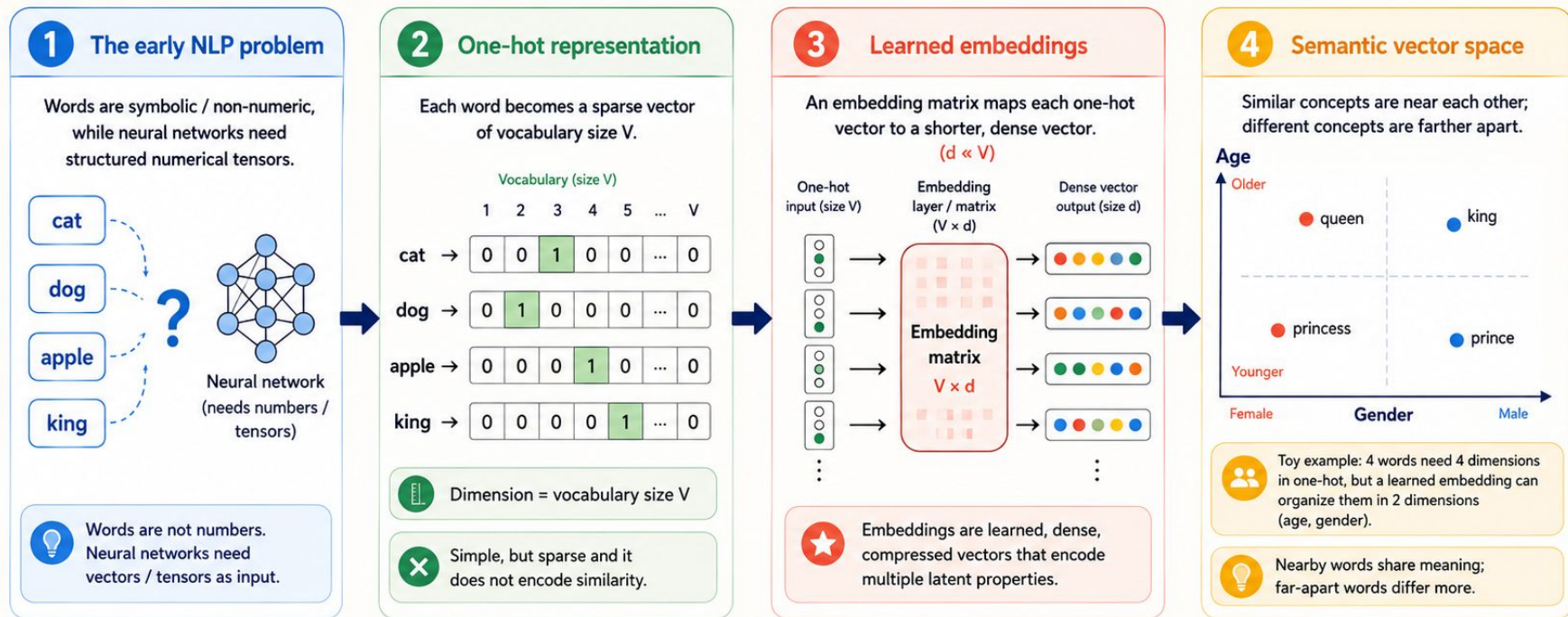


Backgrounds

What is Embedding?

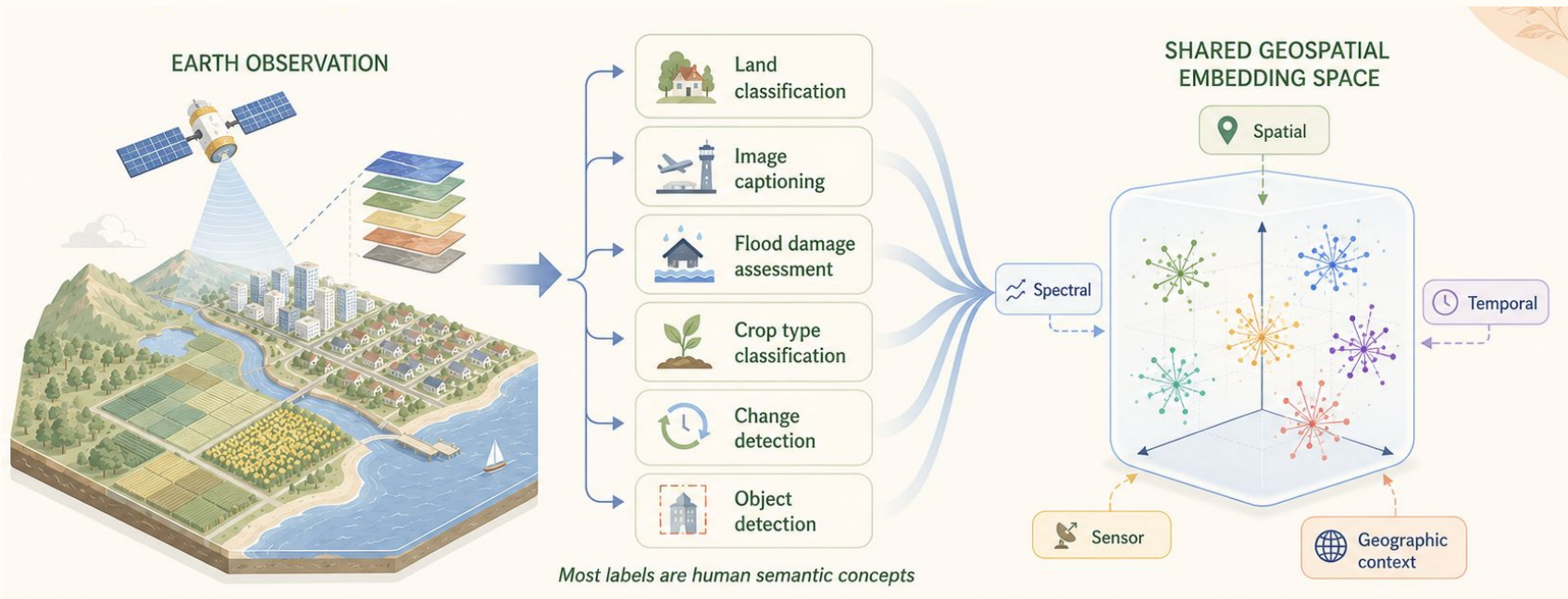
Why Geospatial needs Embeddings?

What is Embedding?

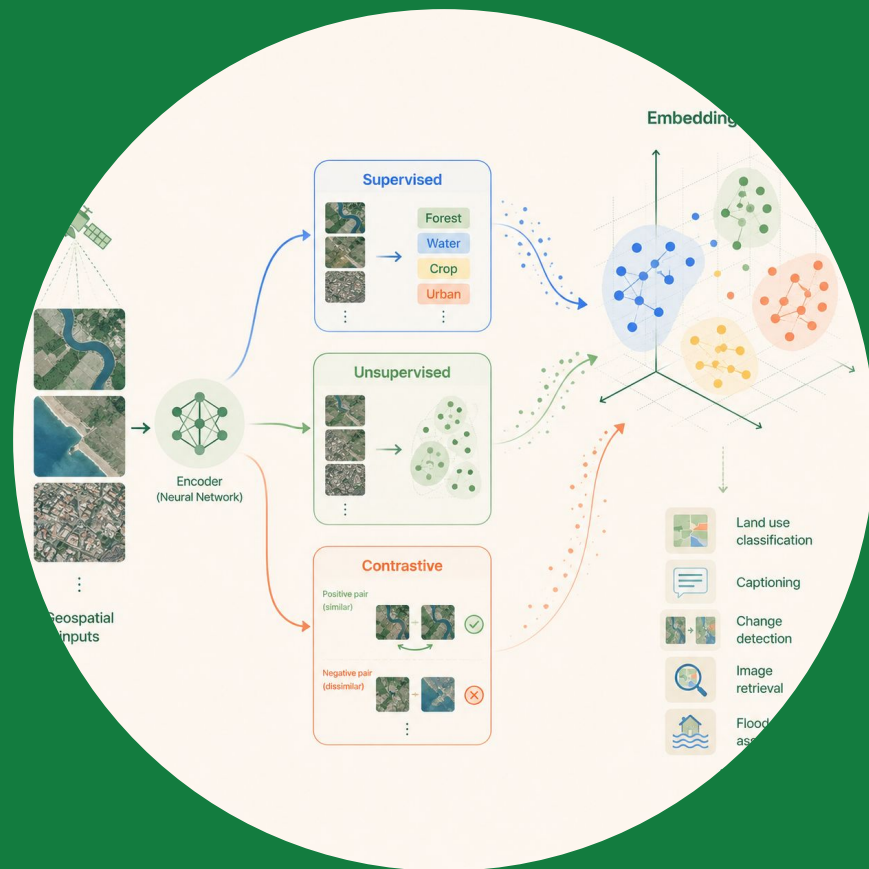


Embeddings: Compressing semantics into low-dimension of vectors.

Why Geospatial Needs Embeddings?



Like pixels for human, embedding is a representation for machines to understand image pixels, time series, sensor, location, and semantics altogether.



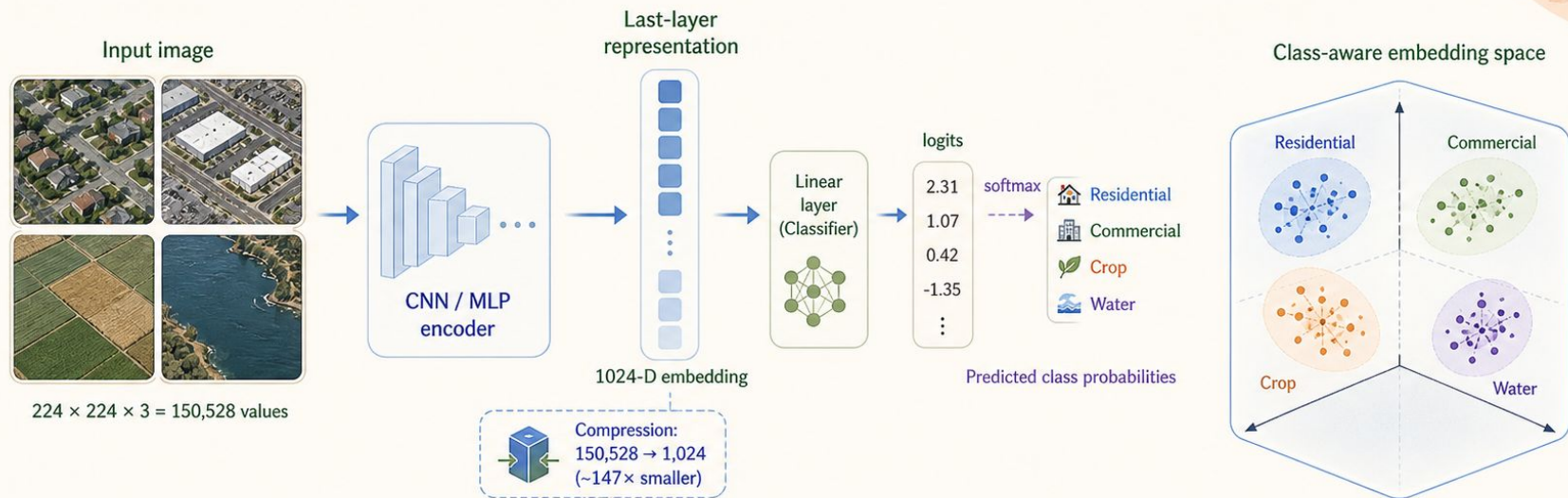
How to Generate Embeddings?

Supervised

Unsupervised

Contrastive

Embedding Generation: Supervised Learning



Although not commonly defined as so, traditional classification models' last layers generate "embeddings" for each image, which contains the pre-defined categorical semantics.

Hands on Time - Supervised Learning (10 mins)

EuroSAT pretrained classifier.

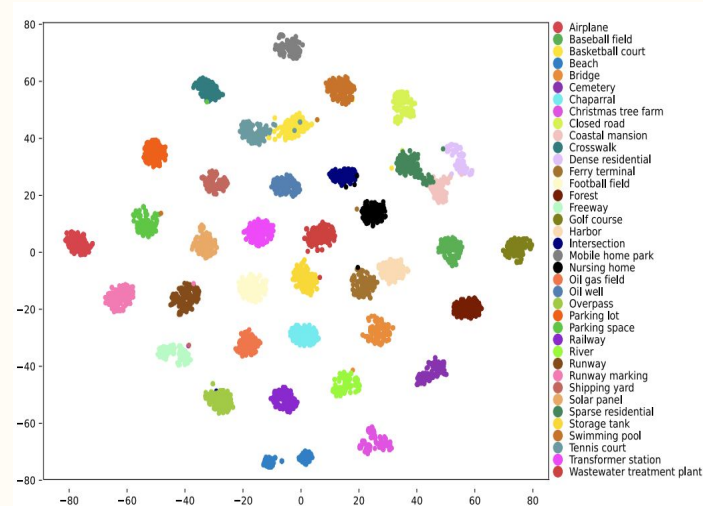
Data: <https://github.com/pheelber/EuroSAT>

Model: <https://huggingface.co/cm93/resnet50-eurosat>

Generate the last layer embeddings on eurosat test set.

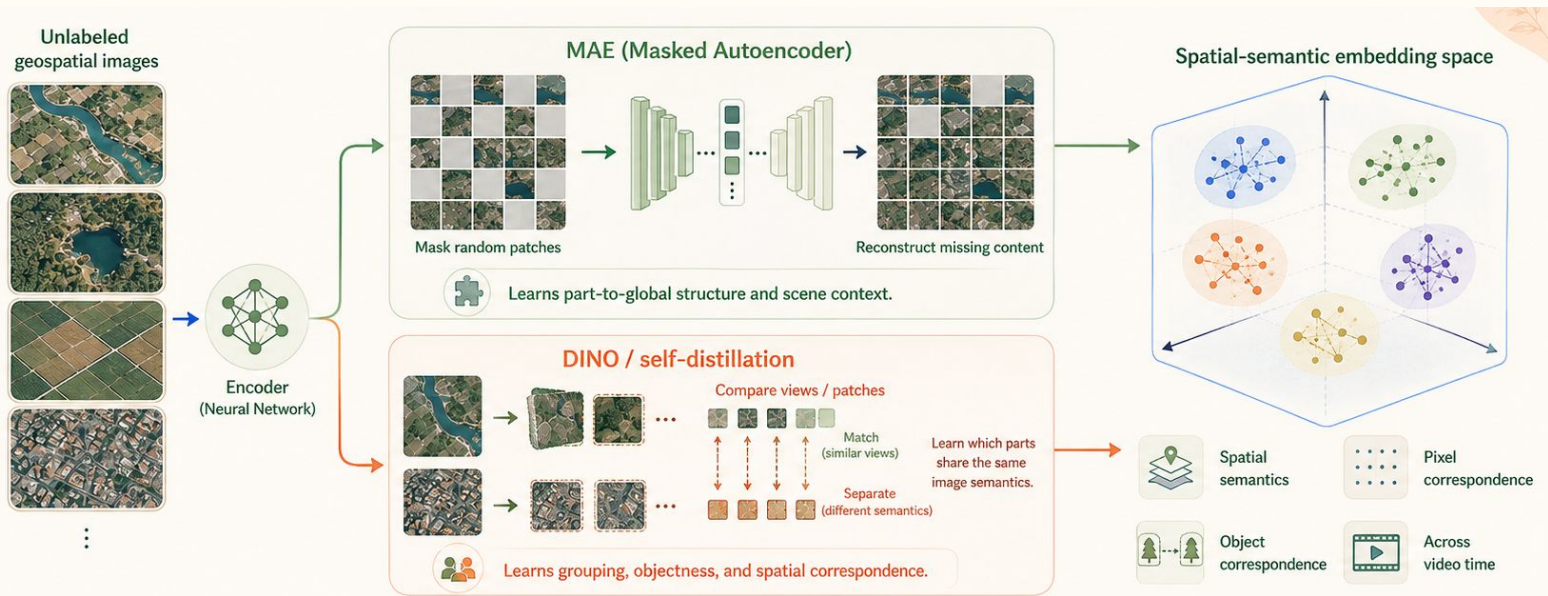
Run 2D T-SNE visualization.

[Notebook#0]



Run and see the 2D t-SNE.

Embedding Generation: Unsupervised Learning



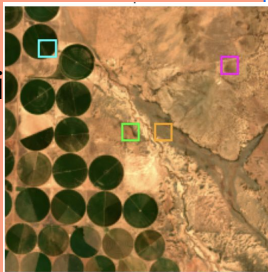
By learning to “reconstruct” (MAE) or “differentiate” (DINO), such foundational models learn the semantic similarities across either spatial pixels or temporal changes.

Hands on Time - Unsupervised Learning (10 mins)

MAE spatial similarity
of some Geospatial Images
(compare image patch similarity)

[Notebook#1]

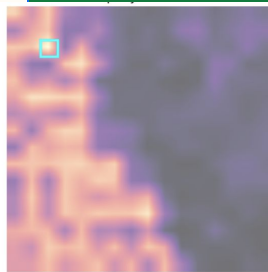
IBM-NASA Prithi



DINOv3 spatial similarity
of same Geospatial Images
(compare image patch similarity)

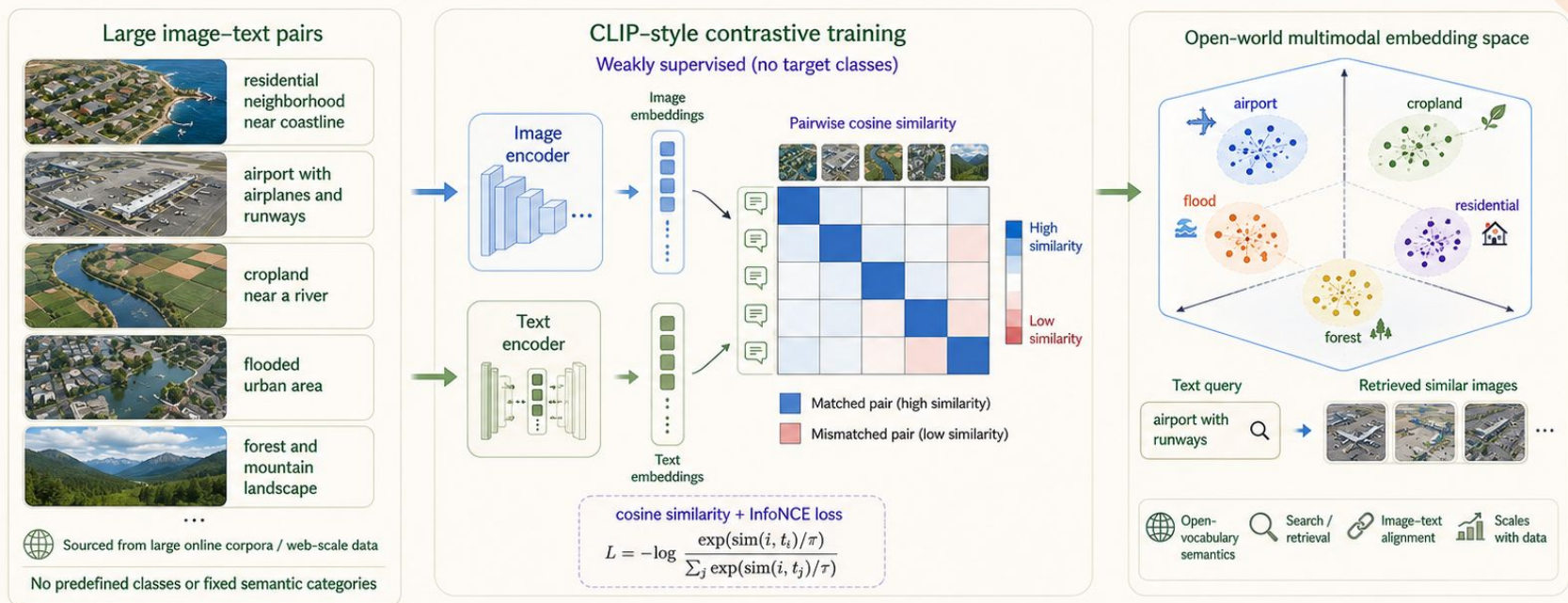
[Notebook#2]

META DINOv3



Run and see the difference.

Embedding Generation: Contrastive Learning



CLIP (contrastive language image pretraining) for the first-time shows the power of “embedding” - how one compressed vector from the image can contain various semantics.

Hands on Time - Contrastive Learning (10 mins)

Text to Image Search

pre-generated embeddings + raw image

⇒ Users input language

⇒ Fine target image

[Notebook#3]

Git10m CLIP:

<https://huggingface.co/lcybuaa/Git-RSCLIP-base>

Image to Image Search

embeddings + raw image

⇒ Users input one image

⇒ Find similar image

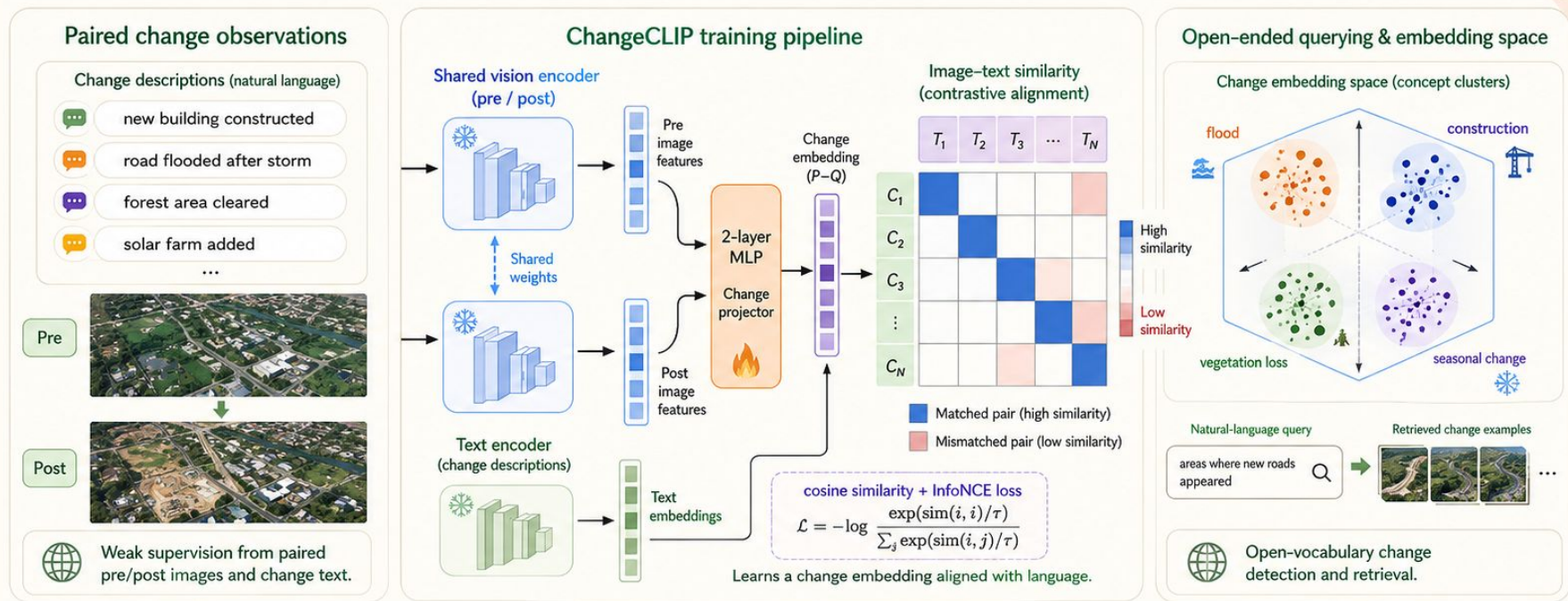
[Notebook#4]

Git10m CLIP:

<https://huggingface.co/lcybuaa/Git-RSCLIP-base>

Run and see the difference.

Embedding Generation: Change-CLIP and more...



Extending CLIP, we can train change detection CLIP, a model that could use natural language to query any type of geospatial change semantics.

Hands on Time - Change CLIP (10 mins)

Text to Image Search

Some pre-generated embeddings + raw image pairs

⇒ Users input language

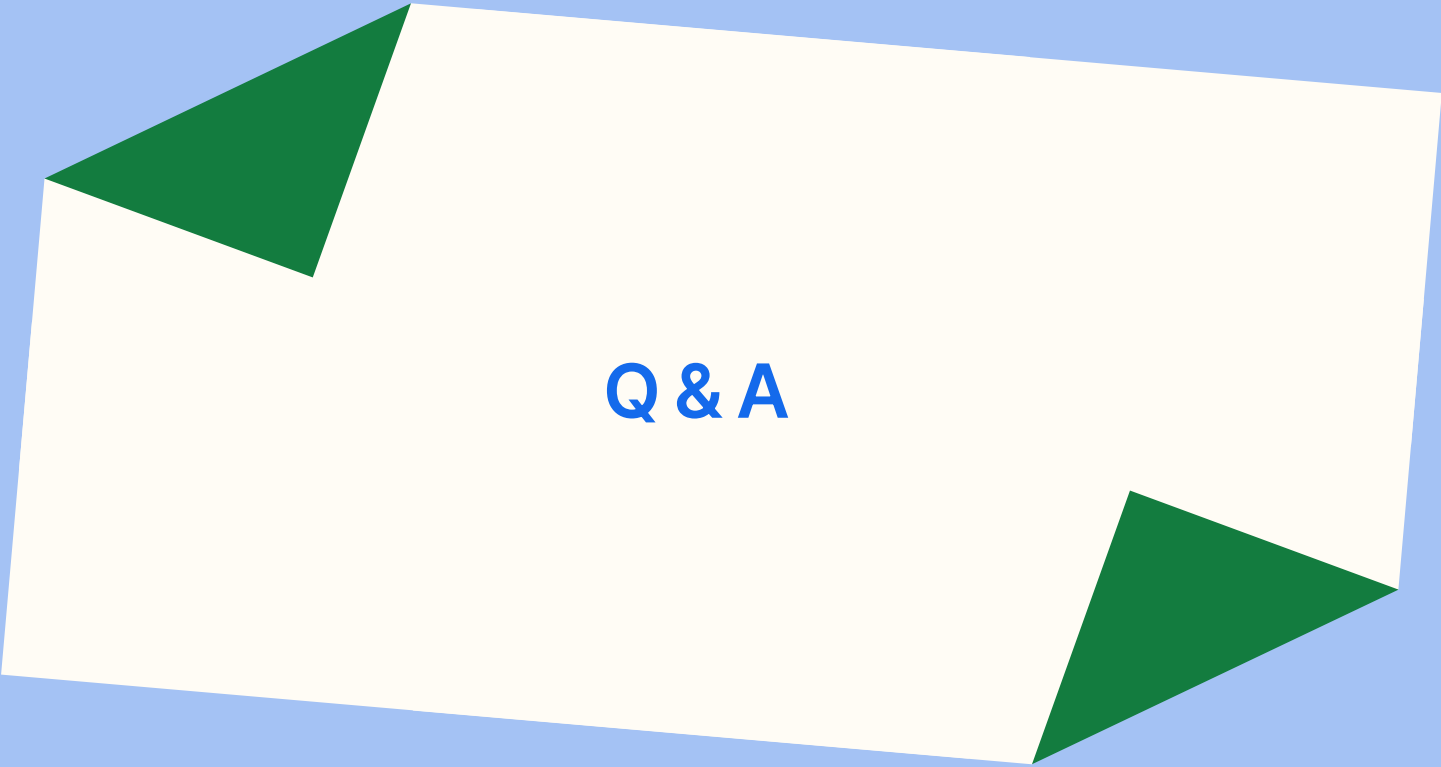
⇒ Fine target change pairs

[Notebook#5 & 6]



Geospatial Embeddings: Understanding Real World in Machine Speed

(Demo - 5 mins)



Q & A